

POINTE AUX BARQUES, MI, USA

WMO: 725386

Lat: 44.022N

Lon: 82.793W

Elev: 590

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94898

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.9	7.4	-11.0	3.1	9.4	-5.5	4.2	11.9	29.8	26.3	26.8	24.7	8.4	230	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	86.3	71.4	82.9	70.3	80.1	69.0	74.6	82.4	72.9	79.9	71.2	77.6	7.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	121.2	79.3	70.3	114.0	77.4	68.8	108.0	75.9	38.4	82.9	36.8	80.0	35.3	77.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	20.6	18.1	DB	-2.1	92.2	6.9	3.1	-7.1	94.4	-11.1	96.2	-15.0	97.9	-20.1	100.2	
			WB	-2.9	77.9	6.5	2.2	-7.6	79.5	-11.4	80.8	-15.0	82.0	-19.7	83.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	24.7	24.4	32.4	42.8	53.5	63.5	70.0	69.3	63.2	52.5	40.4	31.1	(6)
	DBStd	18.09	9.74	9.49	9.76	8.27	8.64	7.57	5.62	5.04	6.82	7.87	8.17	7.77	(7)
	HDD50	3319	785	716	554	250	57	1	0	0	1	63	306	586	(8)
	HDD65	6889	1250	1136	1011	666	375	118	17	15	114	397	739	1051	(9)
	CDD50	2383	0	0	8	35	165	406	619	598	396	139	17	0	(10)
	CDD65	479	0	0	0	1	18	73	171	148	59	9	0	0	(11)
	CDH74	2678	0	0	1	20	151	493	987	701	278	47	0	0	(12)
	CDH80	653	0	0	0	2	39	145	250	144	64	9	0	0	(13)
(14) Wind	WSAvg	7.8	9.3	9.0	8.6	9.5	7.4	5.7	5.5	5.8	6.4	8.2	9.1	9.0	(14)
(15) Precipitation	PrecAvg	38.2	3.0	2.6	2.5	3.4	3.7	3.6	3.4	3.4	3.1	3.5	3.0	2.9	(15)
	PrecMax	46.5	5.2	5.1	5.4	6.9	8.3	5.9	5.2	5.6	6.1	6.0	6.5	6.1	(16)
	PrecMin	31.9	0.7	0.9	0.7	1.3	1.5	1.7	1.4	1.7	0.6	0.7	0.8	1.1	(17)
	PrecStd	3.7	1.2	1.1	1.2	1.3	1.3	1.3	0.9	0.9	1.3	1.4	1.3	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	54.4	66.7	76.8	85.5	89.6	90.5	89.7	86.5	79.9	65.8	54.1	(19)
		MCWB	48.0	49.1	58.0	59.9	67.1	71.6	75.9	73.6	71.6	65.7	57.6	50.8	(20)
	2%	DB	46.3	44.2	58.2	69.2	78.4	84.3	86.0	83.9	81.1	73.0	60.8	48.3	(21)
		MCWB	41.3	38.2	51.6	56.8	64.1	70.2	72.3	71.6	68.2	63.2	53.5	44.2	(22)
	5%	DB	40.4	39.4	50.8	62.1	73.0	80.0	82.4	80.9	76.8	68.4	55.7	44.4	(23)
		MCWB	36.9	34.9	44.5	52.9	61.7	68.4	71.1	70.2	67.3	60.1	50.3	39.4	(24)
	10%	DB	37.1	36.4	44.6	55.1	68.0	75.3	79.4	77.9	73.1	64.0	52.2	40.8	(25)
		MCWB	34.1	32.7	39.5	48.0	59.2	66.0	69.7	68.0	65.8	57.0	47.3	36.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.8	48.4	59.1	62.5	70.8	74.5	77.4	76.6	74.5	69.5	58.6	51.7	(27)
		MCDB	52.3	53.0	65.5	72.5	79.7	85.0	86.8	84.1	82.5	76.2	63.2	54.0	(28)
	2%	WB	42.1	39.6	52.2	57.9	66.9	72.0	75.1	73.8	71.4	64.9	54.2	44.8	(29)
		MCDB	45.3	42.0	57.8	67.0	74.5	81.9	83.0	80.8	76.6	70.6	59.1	47.4	(30)
	5%	WB	37.3	35.4	45.3	53.3	63.6	69.6	73.2	72.2	69.3	61.4	50.8	40.9	(31)
		MCDB	39.5	38.9	50.9	61.7	71.8	77.5	79.9	78.5	74.4	67.0	55.0	43.6	(32)
	10%	WB	34.6	33.0	39.7	48.5	59.8	67.2	71.3	70.4	67.1	58.4	47.4	37.4	(33)
		MCDB	37.0	35.9	44.4	55.0	67.2	74.0	77.3	76.1	72.5	63.4	51.4	40.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.3	12.6	13.3	14.6	14.0	13.9	13.2	13.3	12.1	10.6	8.5	(35)
		MCDBR	17.6	17.9	21.5	26.3	23.1	20.3	18.1	17.1	18.2	19.2	16.5	12.8	(36)
	5% WB	MCWBR	14.8	14.6	16.1	17.3	14.2	11.8	10.1	9.8	10.2	12.1	13.5	11.1	(37)
		MCWBR	15.3	15.7	20.6	24.5	20.6	18.4	15.7	14.6	15.5	16.7	15.5	12.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.305	0.312	0.359	0.383	0.411	0.411	0.403	0.372	0.354	0.318	0.286	(40)
		taud	2.429	2.380	2.429	2.355	2.361	2.307	2.342	2.351	2.427	2.442	2.506	2.516	(41)
	Ebn at Noon	Ebn at Noon	263	282	295	287	282	273	272	270	271	260	255	259	(42)
		Edh at Noon	23	29	32	37	38	40	39	37	32	28	22	20	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	381	645	1099	1456	1726	1927	1928	1652	1302	793	455	297	(44)
	RadStd	RadStd	39	74	113	127	150	116	83	75	122	80	65	33	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (39 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page